



GE Medical Systems

Technical Publication

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Revision 0

Service Note

T-1437, Resistor Pack Failures

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REVISION HISTORY

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Revision	Date	Reason for change
0	12/05/00	Initial Release

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STATEMENT OF PROBLEM

This Service Note is being generated as a result of investigation done into a failure of a group of resistor packs ***manufactured by Bourns***. The failure mode of the resistor pack is a resistive or complete shorting of adjacent IC pins.

The intent of this service note is to provide a listing of circuit boards and locations that use this style of resistor pack. In addition this service note details the test procedure and failure criteria.

As the symptoms caused by a failure of this resistor pack are many, we are not able to create a comprehensive list of the errors or symptoms. If you have a system that is having problems that point to one of the components in the list (see section 1), this document can be used to confirm or deny the failure of the related resistor pack.

IDENTIFICATION AND TESTING

SECTION 1 - IDENTIFYING AFFECTED RESISTOR PACKS

As noted above, the affected resistor packs were manufactured by **Bourns**. If the resistors located in your system are of a different manufacturer you can assume that they are good.

The manufacturers part number of the resistor packs that have had confirmed failures on CT Products is 4116R-1-100. This is a 16 pin Molded DIP package. Figure 1 is an example of how the IC is labeled if it was manufactured by **Bourns**. The affected date code (right of **Bourns** symbol) range is 9800 through 2031.

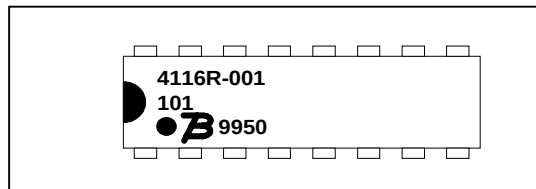


Figure 1 - Bourns Resistor Pack

SECTION 2 - LOCATING RESISTORS

Listed below are the circuit boards that have had confirmed failures identified. As this problem is related to a design flaw in the 4100R series of **Bourns** resistors, other circuit boards may also have this problem.

Board Number	Rev.	Board Name	Location
2143147	1	KV Bd.	R19
2154834	2	MA Bd.	R262
2179860	3	HEMRC Control Bd.	R13
46-264806G2	1	Axial Control Bd.	R48
46-288512G1	13	Gentry I/O Bd.	R19
46-321276G2	0	Collimator II Bd.	R13
46-288854G1	8	R Com Bd.	R58, R65, R70, R76, R82

Table 1 Board and Resistor Location

SECTION 3 - TESTING THE RESISTOR PACK

The board being questioned must have power removed and be out of the system. The resistance measurements can be made on the board with the resistor pack still in circuit.

The part is deemed defective if there is less than 750 K Ohms of resistance measured between adjacent components. This is measured by attaching the leads of an ohm meter to the IC legs (an IC clip will make the job easier). Figure 2 shows the layout of the resistors within the resistor pack.

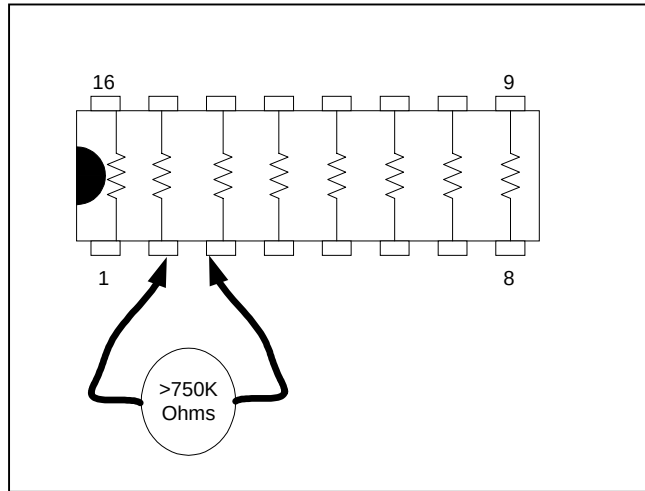


Figure 2- Resistor Pack Circuit Type 001

SECTION 4 - OTHER RESISTOR PACK CONFIGURATIONS

Since the problem found in this resistor pack is due to a design related to the **Bourns 4100R series design**, the following information is included. This information is intended to help identify and test other resistor packs that are used.

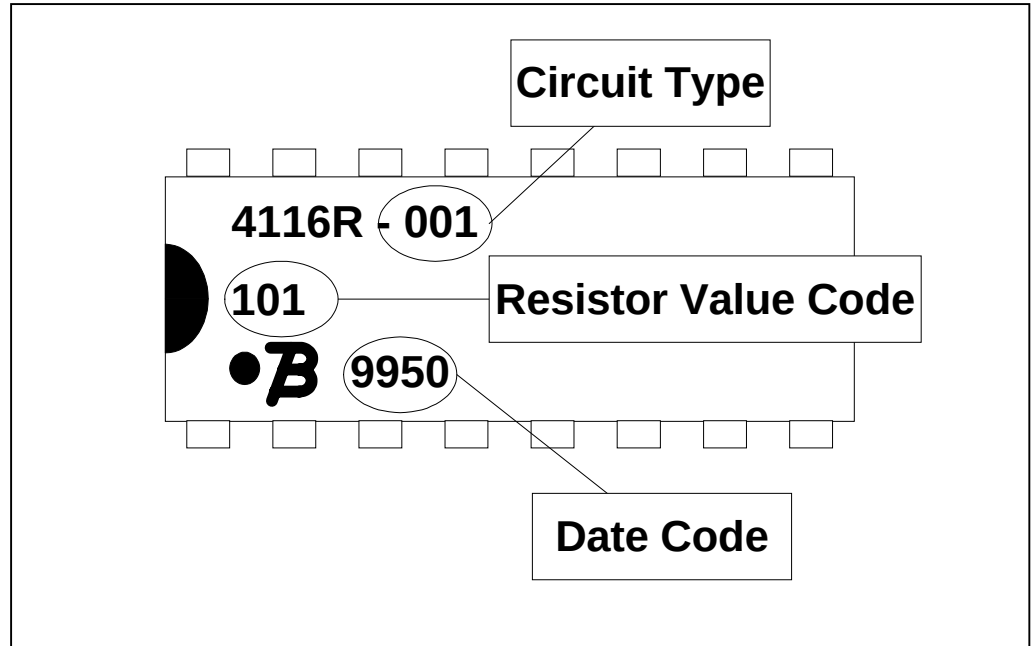


Figure 3- Resistor Pack Labeling

Table 2 is a partial list of the values as they relate to the resistor code found on the IC.

Ohms	Code	Ohms	Code	Ohms	Code	Ohms	Code	Ohms	Code
10	100	470	471	2000	202	8200	822	56000	563
100	101	560	561	2200	222	10000	103	82000	823
150	151	820	821	3300	332	12000	123	100000	104
180	181	1000	102	4700	472	15000	153	150000	154
220	221	1200	122	5600	562	20000	203	470000	474
330	331	1500	152	6800	682	27000	273	1000000	105

Table 2 Resistor Value Code Chart

Figure 4 demonstrates a method of testing the IC configuration used in most pull up circuits. In this figure an Ohm meter is used to verify the value read from the common pin (pin 16) to any other pin. The circuit is considered bad if the reading is less than 90% of the specified value for the given IC.

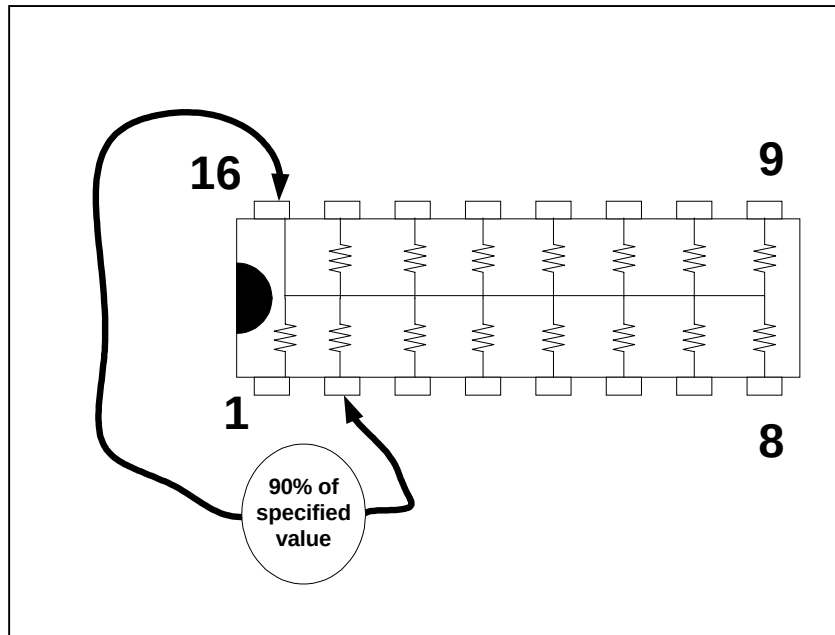


Figure 4- Resistor Pack Circuit Type 002

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